

GLOSSARY

adaptive development Social system evolution by a process of problem solving that includes broad community participation with monitoring to assess the effectiveness of human actions and corrective measures to bring the actions in line with community goals.

agricultural ecosystem See ECOSYSTEM.

Agricultural Revolution The beginning of agriculture about 10,000–12,000 years ago. Domestication and care of plants and animals as sources of food and other materials for human use.

agroforestry Agricultural ecosystems that include trees.

animism Belief that plants, animals and some non-living parts of nature have spirits or souls.

annual Plant that lives only one year or season

autonomous Free of outside control.

biological community All the living organisms (plants, animals, microorganisms) in an ecosystem.

biological control The control of pest organisms by altering their environment or introducing natural enemies such as predators or pathogens.

biological production (also called 'primary production') Total plant growth (due to photosynthesis) in an ecosystem. The quantity of plant growth determines the food supply for all other living organisms in an ecosystem.

biome A large-scale ecosystem associated with a particular climatic region.

biophilia An inborn emotional need of humans to have plants and animals as a part of their lives in one way or another.

brackish water A mixture of fresh water and salt water.

canopy The top layer of branches and leaves in a forest.

carrying capacity The maximum population number of a particular plant or animal species that an ecosystem can support on a long-term basis.

216 *Human Ecology*

chaparral A biological community composed of shrubby plants adapted to dry summers and moist winters. Common in coastal Southern California.

clear-cutting Cutting all of the trees in a forest at the same time. The opposite of SELECTIVE LOGGING.

climax community The final stage of ecological succession.

coadaptation Adjustment of different parts of an ecosystem to one another.

coevolution Associated changes in two species of living organisms that have a close ecological relationship (eg, predator/prey), acting as agents of natural selection for one another.

commons A tract of land or other resource used jointly by the members of a community.

community assembly Self-organization of biological communities by selective addition of new species of plants or animals that arrive to an ecosystem.

complex adaptive systems Systems with feedback loops that enable them to adjust to fluctuations in their environment in ways that promote their survival.

consumer Animal or other living organism that feeds on plants, animals or microorganisms.

consumption The movement of organic matter (ie, carbon chains) through a food web as animals and microorganisms eat (or otherwise ingest) plants, animals or microorganisms to obtain the material and energy that they need to sustain their lives.

controlled burning Small fires set deliberately to reduce combustible material in a forest.

counterintuitive Opposite or contrary to expectations. The outcomes of human actions in complex adaptive systems such as ecosystems and social systems are often counterintuitive because complex chains of effects generate ultimate consequences that are different from immediate impacts.

decomposer A microorganism that feeds on dead plants, animals or microorganisms.

decomposition Consumption of dead plants, animals or microorganisms by microorganisms.

denial (cognitive dissonance) Refusal to believe information that conflicts with an existing belief system. Denial is a defence mechanism for reducing anxiety due to a conflict between reality and existing beliefs.

desertification Transformation of other kinds of ecosystems (eg, grassland) to desert. Typically associated with loss of topsoil and consequent reduction of plant life in semi-arid regions.

diminishing returns A benefit that beyond a certain point fails to increase in proportion to additional investments.

dispersal The spreading of plants, animals or microorganisms from one place to another by their own movement or when carried by wind, water, animals or machines.

division of labour Diversification of tasks or occupational roles in a society in order to improve working efficiency.

drift net Large-mesh monofilament nylon gillnet, typically miles long, used for ocean fishing. Fish are caught when they become entangled while trying to swim through the net.

ecological competition Use of the same resource by two different species of plant, animal or microorganism.

ecological niche The role of a particular species in the ecosystem. Ecological niche is defined in terms of the physical conditions and resources necessary for the species' survival and the species' position in the ecosystem's foodweb.

ecological succession A systematic progression of biological communities through time, each biological community replacing another due to natural ecological processes. See also HUMAN-INDUCED SUCCESSION.

ecology The science of relationships and interactions between living organisms and their environment.

economy of scale Reduction in unit costs as a consequence of increase in the scale of production.

ecosystem A system formed by the interaction of a biological community with its chemical and physical environment. An ecosystem includes everything at a particular location: plants, animals, microorganisms, air, water, soil and human-built structures. *Natural* ecosystems are formed entirely by natural processes. *Agricultural* ecosystems are created by people to provide food or other materials. *Urban* ecosystems are dominated by human-built structures.

ecosystem inputs Materials, energy or information that move into an ecosystem. *Human inputs* are human activities to organize or structure ecosystems.

ecosystem outputs Materials, energy or information that moves out of an ecosystem to another ecosystem or the human social system.

ecosystem services Materials, energy or information that people obtain from ecosystems for survival (eg, food, fibres, construction materials and water) or as amenities and experiences to enrich their lives.

ecosystem state Particular physical conditions, chemical concentrations, and numbers of each kind of plant, animal and microorganism that characterize an ecosystem at a particular place and time.

218 *Human Ecology*

emergent property A characteristic of a system as a whole that comes into existence from the organization of the system's parts rather than from characteristics of any of the parts themselves.

endangered species A species of plant or animal in danger of extinction, typically as a consequence of human activities.

energy flow Movement of energy in the carbon chains of organic matter that passes through a food web as one organism consumes another.

environmental refugees People who move from a region because the ecosystem is no longer able to provide for their basic needs.

estuary The wide lower course of a river where its currents meet ocean tides. Much of the water in an estuary is a tidal mixture of fresh water and salt water.

eutrophication Pollution of water with minerals that stimulate plant growth.

exponential population growth Increase in population characterized by an increasingly larger population growth rate as the number of individuals in the population increases.

extended economic zone A marine area for which a nation claims sovereignty over all resources for a distance of 320 kilometres from its shores.

fallow Land that is left unused, without ploughing, planting or raising crops.

fisheries succession A change in the biological community of a fisheries ecosystem in which fish species that are intensely harvested disappear and other fish species (or other kinds of animals) take their place.

food chain A series of living organisms connected by one eating another. See also **FOOD WEB**.

food chain efficiency The percentage of carbon-chain energy at one step of a food chain that is available for consumption by the next step of the food chain.

food web A set of interconnected **FOOD CHAINS** which includes all the organisms in an ecosystem's biological community.

Green Revolution Increase in agricultural production through the introduction of high-yield crop varieties and application of modern agricultural techniques.

habitat The type of ecosystem in which a particular kind of plant, animal or microorganism normally lives.

hierarchical organization Organization of a system in such a way that each element of the system contains other elements within it. Biological systems have a hierarchy that extends from atoms and molecules to cells, tissues, organs, individuals, populations and biological communities. Landscape mosaics

have a nested hierarchy of ecosystems extending from less than a square metre to the entire planet Earth.

high-yield varieties Genetically improved crops produced by modern breeding methods to have a high level of production under ideal environmental conditions.

homeostasis Negative feedback that maintains a living organism's body function within limits essential for the body to continue functioning properly despite external stimuli that have a tendency to disrupt the function.

human ecology The science of relationships and interactions between people and their environment.

human-induced succession A change in an ecosystem's biological community as a consequence of human activities. See also ECOLOGICAL SUCCESSION.

Industrial Revolution Changes in economic and social organization that began about 300 years ago in England with replacement of hand tools by power-driven machines.

land subsidence Sinking of the land due to processes such as organic matter decomposition and sediment compaction resulting from the weight of overlying sediments.

landscape mosaic A repetitive patchwork of different kinds of ecosystems across a land area. See also HIERARCHICAL ORGANIZATION.

larvicide A chemical or other agent for killing insect larvae.

leaf litter A layer of dead plant material on the soil surface.

legume Plants such as peas and beans with pods that split along both sides. Legumes commonly have root nodules with symbiotic bacteria that convert atmospheric nitrogen to a form that plants can use.

marsh A low-lying area, saturated with water, all or part of which is typically covered with moisture-tolerant grasses.

material cycling (also called 'nutrient cycling' or 'mineral cycling') Circulation of chemical elements through the food web, air, soil and water in an ecosystem.

metropolitan region A large city and its surrounding suburbs.

Minamata disease A severe form of mercury poisoning characterized by neurological degeneration. Called Minamata because of mercury poisoning from contaminated fish in Minamata Bay, Japan.

mineral cycling See MATERIAL CYCLING.

mineral nutrients Inorganic substances (eg, nitrogen, phosphorous, potassium, calcium, magnesium, sulphur, cobalt, copper, boron, manganese and zinc) that plants need for their growth.

monoculture An agricultural ecosystem with only one kind of crop.

mycorrhizae Fungi in symbiotic association with plant roots, facilitating phosphorous uptake by the roots.

natural capital All the natural resources on which a civilization depends to create economic prosperity. Natural capital includes water, minerals, air, soil, plants, animals and microorganisms in natural, agricultural and urban ecosystems.

natural ecosystem See ECOSYSTEM.

negative feedback A chain of effects through an ecosystem or social system that tends to keep particular parts of the system within certain limits.

nitrogen-fixing bacteria Bacteria responsible for nitrogen fixation – the conversion of atmospheric nitrogen to a form (eg, ammonia) that plants can use.

non-renewable natural resources Non-living resources such as petroleum, gas, coal and minerals.

nutrient cycling See MATERIAL CYCLING.

nutrient pump An ecological process in which trees take up mineral nutrients from soil too deep for crop roots to reach. The mineral nutrients pass into the leaves of trees, eventually falling onto the soil where they are accessible to crops.

organic farming Farming style that uses fertilizers of plant or animal origin and natural pest control methods instead of chemical fertilizers, pesticides, or growth stimulants.

overexploitation Use of an ecosystem service in excess of what the ecosystem can sustain on a long-term basis.

overfishing The OVEREXPLOITATION of a fishery by harvesting more fish than the fishery can yield on a sustainable basis.

overgrazing The OVEREXPLOITATION of pasture or rangeland resources by grazing more livestock than the grasses can sustain.

overshoot To pass beyond. In human ecology overshoot refers to (a) the increase of an animal or plant population beyond the carrying capacity of its environment or (b) the increase of industrial or other demands on an ecosystem beyond the ecosystem's capacity to provide services satisfying the demands.

parasite An animal that obtains its nutrition by living in close association with another kind of animal (the *host*) without killing it immediately. The host animal may be injured (and in some cases eventually killed) by the relationship.

pathogen A microorganism that causes disease in another kind of organism. This normally happens when a pathogen lives in close association with a host organism to procure the habitat and nutrition it requires for survival.

perception The way that people 'see' and interpret information. Perceptions are important for human ecology because they shape the way that information is used to agree on human actions.

perennial Crop or other plant that is present throughout the year because it lives for at least several years.

phytoplankton Microscopic plants that drift in the water of an aquatic ecosystem.

polyculture A mixture of crop species in an agricultural ecosystem.

population All the plants, animals or microorganisms of the same species in a particular ecosystem.

population pressure Stress due to scarcity of food or other resources when a population is close to, or greater than, carrying capacity.

population regulation Control of population number by negative feedback.

positive feedback A chain of effects through an ecosystem or social system that amplifies change.

precautionary principle A standard for human/environment interaction that emphasizes prudent action due to limited knowledge of the environment.

predator An animal that eats other animals.

primary production See BIOLOGICAL PRODUCTION.

redundancy Duplication or overlap of function that exceeds what is necessary.

renewable natural resources Resources that are continually replaced by material cycling and energy flow in an ecosystem. Most renewable resources (eg, forests, fisheries and agricultural products) are living resources, though some non-living resources (eg, water) are also renewable.

resilience The ability to return to an original form after severe stress or disturbance.

respiration Oxidation of carbon chains in the body of living organisms to extract energy for metabolic processes.

salinization Accumulation of toxic concentrations of salts in the soil as a consequence of irrigation. Irrigation water evaporates from the field, leaving dissolved salts to accumulate in the soil.

salt-water intrusion Tidal movement of ocean water inland because river flow to the ocean is interrupted.

satoyama A traditional system of village agriculture and forest management in Japan.

selective logging Cutting only some trees in a forest. Selective logging is a way to manage a forest on a sustainable basis. The opposite of CLEAR-CUTTING.

shifting cultivation See SWIDDEN.

slash-and-burn agriculture See SWIDDEN.

social institutions An established pattern of behaviour or relationships accepted as a fundamental part of a culture.

222 *Human Ecology*

social organization The structure of social relations within a group, including relations among different subgroups and institutions.

social system Everything about human society, including its organization and structure, knowledge and technology, language, culture, perceptions and values.

soil erosion Loss of soil that is worn or carried away by wind or rain.

stability Constancy. Resistance to change.

stability domain A set of similar system states characterized by natural or social processes that tend to keep the system in those states.

subsistence farming Farming whose products provide basic family needs with little surplus for marketing.

supply zone See ZONE OF INFLUENCE.

sustainable development Doing things in a way that does not reduce the opportunities of future generations to meet their needs. *Ecologically sustainable development* depends upon human–ecosystem interaction that maintains the functional integrity of ecosystems in a way that allows them to continue providing ecosystem services.

swamp A forest ecosystem that is saturated with water.

swidden (also called ‘slash-and-burn agriculture’ or ‘shifting cultivation’) An agricultural system characterized by rotation between crops and natural vegetation. Fields are prepared for cultivation by cutting and burning natural vegetation (eg, forest). The field typically has crops for one to three years, after which it is left to fallow, generating natural vegetation that is eventually burned to prepare the field once again for crops.

symbiosis Mutually beneficial association between two different species of organisms.

tragedy of the commons The OVEREXPLOITATION of a natural resource because no one in particular owns the resource, resource use is open to anyone without restriction, and the resource is large enough for the actions of single individuals to have no significant effect on the supply of the resource.

trash fish Fish that have little commercial value.

unsustainable Not able to be continued for a long period. *Ecologically unsustainable* refers to human–ecosystem interaction that damages an ecosystem or depletes a resource in a way that diminishes the supply of the resource or the capacity of the ecosystem to provide a service.

urban ecosystem See ECOSYSTEM.

values Emotionally respected ideals, customs and institutions of a society.

vector An animal that transmits bacterial, viral, fungal or other disease.

watershed A region or area that drains to a stream, river, lake or ocean. Watersheds are the main source of water for cities and irrigated agriculture.

wetland A lowland area, such as a marsh or swamp, that is saturated with moisture.

worldview A person's comprehensive conception or image of the surrounding world and his relation to it. Worldviews shared generally by everyone in a society constitute the society's worldview.

zone of influence The area surrounding a city and affected by the city's authority or commerce.

zooplankton Small animals that live in the water of an aquatic ecosystem.

FURTHER READING

HUMAN ECOLOGY

- Anderson, E (1996) *Ecologies of the Heart: Emotion, Belief, and the Environment*, Oxford University Press, Oxford
- Borden, R, Jacobs, J and Young, G (eds) (1986) *Human Ecology: A Gathering of Perspectives*, Society for Human Ecology, College Park, Maryland
- Borden, R, Jacobs, J and Young, G (eds) (1988) *Human Ecology: Research and Applications*, Society for Human Ecology, College Park, Maryland
- Botkin, D and Keller, E (1999) *Environmental Science: Earth as a Living Planet*, Wiley, New York
- Boyden, S (1987) *Western Civilization in Biological Perspective: Patterns in Biohistory*, Clarendon, Oxford
- Brown, L R, Flavin, C and French, H (2001) *State of the World 2001: A Worldwatch Institute Report on Progress Toward a Sustainable Society*, Earthscan, London
- Brown, L R, Flavin, C, French, H, Postel, S and Starke, L (2000) *State of the World 2000: A Worldwatch Institute Report on Progress Toward a Sustainable Society*, Earthscan, London
- Brown, L R, Flavin, C, French, H, and Starke, L (1999) *State of the World 1999: A Worldwatch Institute Report on Progress Toward a Sustainable Society*, Earthscan, London
- Bunyard, P (1996) *Gaia in Action: Science of the Living Earth*, Floris Books, Edinburgh
- Campbell, G (1995) *Human Ecology: The Story of our Place in Nature from Prehistory to Present*, Aldine, Hawthorne
- Clayton, A, and Radcliffe, N (1996) *Sustainability: A Systems Approach*, Earthscan, London
- Cohen, J (1995) *How Many People can the Earth Support?*, WW Norton, New York
- Daily, G (1997) *Nature's Services: Societal Dependence on Ecosystems*, Island Press, Washington, DC

- Diamond, J (1997) *Guns, Germs, and Steel: The Fates of Human Societies*, WW Norton, New York
- Drury, W and Anderson, J (1998) *Chance and Change: Ecology for Conservationists*, University of California Press, Berkeley
- Ehrlich, P, Ehrlich, A and Holdren, J (1973) *Human Ecology: Problems and Solutions*, W H Freeman, San Francisco
- Foreman, R (1995) *Land Mosaics: The Ecology of Landscapes and Regions*, Cambridge University Press, Cambridge
- Gliessman, S (1997) *Agroecology: Ecological Processes in Sustainable Agriculture*, Lewis Publishers, Boca Raton
- Gunderson, L, Holling, C and Light, S (1995) *Barriers and Bridges to Renewal of Ecosystems and Institutions*, Columbia University Press, New York
- Hardin, G (1993) *Living within Limits: Ecology, Economics and Population Taboos*, Oxford University Press, Oxford
- Hawken, P, Lovins, A B and Lovins L (1999) *Natural Capitalism: Creating the Next Industrial Revolution*, Earthscan, London
- Hawley, A (1950) *Human Ecology: A Theory of Community Structure*, Ronald, New York
- Hens, L, Borden, R and Suzuki, S (1999) *Research in Human Ecology: An Interdisciplinary Overview*, Vu University Press, Amsterdam
- Holling, C (1978) *Adaptive Resource Management and Assessment*, Wiley-Interscience, New York
- Homer-Dixon, T (1999) *Environment, Scarcity, and Violence*, Princeton University Press, Princeton
- Hrdy, S (1999) *Mother Nature: A History of Mothers, Infants, and Natural Selection*, Pantheon, New York
- Karliner, J (1997) *The Corporate Planet: Ecology and Politics in the Age of Globalization*, Sierra Club Books, San Francisco
- Kormondy, E and Brown, D (1998) *Fundamentals of Human Ecology*, Prentice-Hall, New York
- Lee, K (1993) *Compass and Gyroscope*, Island Press, Washington, DC
- Levin, S (1999) *Fragile Dominion: Complexity and the Commons*, Perseus, Reading, Massachusetts
- Lewin, R (1992) *Complexity: Life at the Edge of Chaos*, Macmillan, New York
- Lovelock, J (1979) *Gaia: A New Look at Life on Earth*, Oxford University Press, Oxford
- Marten, G (1986) *Traditional Agriculture in Southeast Asia: A Human Ecology Perspective*, Westview, Boulder, Colorado
- McHarg, I (1995) *Design with Nature*, Wiley, New York
- Meadows, D, Meadows, D and Randers, J (1993) *Beyond the Limits: Confronting Global Collapse, Envisioning a Sustainable Future*, Chelsea Green, Post Mills, Vermont
- Miller, G (1998) *Environmental Science: Working with the Earth*, Wadsworth, Belmont, California
- Moran, E (1982) *Human Adaptability: An Introduction to Ecological Anthropology*, Westview, Boulder, Colorado

226 Human Ecology

- Nebel, B and Wright R (1999) *Environmental Science: The Way the World Works*, Prentice-Hall, New York
- Norgaard, R (1994) *Development Betrayed: The End of Progress and a Coevolutionary Revisioning of the Future*, Routledge, New York
- Ostrom, E (1990) *Governing the Commons: The Evolution of Institutions for Collective Action*, Cambridge University Press, Cambridge
- Rambo, A and Sajise, T (1985) *An Introduction to Human Ecology Research on Agricultural Systems in Southeast Asia*, University of the Philippines, Los Banos, Philippines
- Rees, W, Testemale, P and Wackernagel, M (1995) *Our Ecological Footprint: Reducing Human Impact on the Earth*, New Society Publishers, Gabriola Island, British Columbia
- Roseland, M (1997) *Eco-City Dimension: Healthy Communities, Healthy Planet*, New Society Publishers, Gabriola Island, British Columbia
- Soule, F and Piper, J (1992) *Farming in Nature's Image: An Ecological Approach to Agriculture*, Island Press, Washington, DC
- Spirn, A (1999) *Language of Landscape*, Yale University Press, New Haven
- Steele, J (1997) *Sustainable Architecture: Principles, Paradigms, and Case Studies*, McGraw-Hill, New York
- Suzuki, S, Borden, R and Hens, L (eds) (1991) *Human Ecology – Coming of Age: An International Overview*, VUB Press, Brussels
- Tainter, J (1990) *Collapse of Complex Societies*, Cambridge University Press, Cambridge
- Trefil, J (1994) *A Scientist in the City*, Doubleday, New York
- Troxel, J (1994) *Participation Works: Business Examples from Around the World*, Miles Rivers Press, Alexandria, Virginia
- United Nations Development Programme, United Nations Environment Programme, World Bank and World Resources Institute (2000) *People and Ecosystems: The Fraying Web of Life*, Elsevier, New York
- Van der Ryan, S and Cowan, S (1996) *Ecological Design*, Island Press, Washington, DC
- Watt, K (1973) *Principles of Environmental Science*, McGraw-Hill, New York
- Watt, K (2000) *Encyclopedia of Human Ecology: New Approaches to Understanding Societal Problems*, Academic Press, New York
- Wenn, D (1996) *Deep Design: Pathways to a Livable Future*, Island Press, Washington, DC
- Wilson, E (2000) *Sociobiology: The New Synthesis*, Harvard University Press, Cambridge
- Weeks, W (1997) *Beyond the Ark: Tools for an Ecosystem Approach to Conservation*, Island Press, Washington, DC
- World Commission on Environment and Development (1987) *Our Common Future*, Oxford Paperbacks, Oxford
- World Resources Institute (1997) *Frontiers of Sustainability*, Island Press, Washington, DC
- Yes! A Journal for Positive Futures, PO Box 10818, Bainbridge Island, Washington 98110, USA

DENGUE HEMORRHAGIC FEVER, MOSQUITOES AND COPEPODS

- Brown, M, Kay, B and Hendrix, J (1991) 'Evaluation of Australian *Mesocyclops* (Copepoda: Cyclopoida) for mosquito control', *Journal of Medical Entomology*, vol 28, pp618–623
- Christophers, S (1960) *Aedes Aegypti* (L.). *The Yellow Fever Mosquito: Its Life History, Bionomics and Structure*, Cambridge University Press, Cambridge
- Focks, D, Haile, D, Daniels, E and Mount, G (1993) 'Dynamic life table model for *Aedes aegypti* (Diptera: Culicidae): analysis of the literature and model development', *Journal of Medical Entomology*, vol 30, pp1003–1017
- Halstead, S (1997) 'Epidemiology of dengue and dengue hemorrhagic fever' in Gubler, D and Kuno, G (eds) *Dengue and Dengue Hemorrhagic Fever*, CAB International, New York
- Halstead, S (1998) 'Dengue and dengue hemorrhagic fever' in Feigin, R and Cherry, J (eds) *Textbook of Pediatric Infectious Diseases*, W B Sanders, Philadelphia
- Halstead, S and Gomez-Dantes, H (eds) (1992) *Dengue – a worldwide problem, a common strategy, Proceedings of an International Conference on Dengue and Aedes aegypti Community-based Control*, Mexican Ministry of Health and Rockefeller Foundation, Mexico
- Marten, G (1984) 'Impact of the copepod *Mesocyclops leuckarti pilosa* and the green alga *Kirchneriella irregularis* upon larval *Aedes albopictus* (Diptera: Culicidae)', *Bulletin of the Society for Vector Ecology*, vol 9, pp1–5
- Marten, G, Astaeza, R, Suárez, M, Monje, C and Reid, J (1989) 'Natural control of larval *Anopheles albimanus* (Diptera: Culicidae) by the predator *Mesocyclops* (Copepoda: Cyclopoida)', *Journal of Medical Entomology*, vol 26, pp624–627
- Marten, G (1990) 'Evaluation of cyclopoid copepods for *Aedes albopictus* control in tires', *Journal of American Mosquito Control Association*, vol 6, pp681–688
- Marten, G (1990) 'Elimination of *Aedes albopictus* from tire piles by introducing *Macrocyclus albidus* (Copepoda, Cyclopoida)', *Journal of American Mosquito Control Association*, vol 6, pp689–693
- Marten, G, Bordes, E and Nguyen, M (1994) 'Use of cyclopoid copepods for mosquito control', *Hydrobiologia*, vol 292/293, pp491–496
- Marten, G, Borjas, G, Cush, M, Fernández, E, and Reid, J (1994) 'Control of larval *Ae. aegypti* (Diptera: Culicidae) by cyclopoid copepods in peridomestic breeding containers', *Journal of Medical Entomology*, vol 31, pp36–44
- Marten, G, Thompson, G, Nguyen, M, and Bordes, E (1997) *Copepod Production and Application for Mosquito Control*, New Orleans Mosquito Control Board, New Orleans, Louisiana

- Riviere, F and Thirel, R (1981) 'La predation du copepods *Mesocyclops leuckarti pilosa* sur les larves de *Aedes (Stegomyia) aegypti* et *Ae. (St.) polynesiensis* essais preliminaires d'utilisation comme de lutte biologique', *Entomophaga*, vol 26, pp427-439
- Nam, V, Yen, N, Kay, B, Marten, G and Reid, J (1998) 'Eradication of *Aedes aegypti* from a village in Vietnam, using copepods and community participation', *American Journal of Tropical Medicine and Hygiene*, vol 59, pp657-660
- Soper, F, Wilson, D, Lima, S and Antunes W (1943) *The Organization of Permanent Nation-wide anti-Aedes aegypti Measures in Brazil*, The Rockefeller Foundation, New York
- Suarez, M, Ayala, D, Nelson, M and Reid, J (1984) 'Hallazgo de *Mesocyclops aspericornis* (Daday) (Copepoda: Cyclopoida) depredador de larvas de *Aedes aegypti* en Anapoima-Colombia', *Biomedica*, vol 4, pp74-76

BARATARIA-TERREBONNE NATIONAL ESTUARY PROGRAM

- BTNEP (1995) *Land Use and Socioeconomic Status and Trends in the Barataria-Terrebonne Estuarine System*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- BTNEP (1995) *Saving Our Good Earth: A Call to Action. Barataria-Terrebonne estuarine system characterization report*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- BTNEP (1995) *Status and Trends of Eutrophication, Pathogen Contamination, and Toxic Substances in the Barataria-Terrebonne Estuarine System*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- BTNEP (1995) *Status and Trends of Hydrologic Modification, Reduction in Sediment Availability, and Habitat Loss/Modification in the Barataria-Terrebonne Estuarine System*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- BTNEP (1995) *Status, Trends, and Probable Causes of Change in Living Resources in the Barataria-Terrebonne Estuarine System*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- BTNEP (1996) *The Estuary Compact: A Public-Private Promise to Work Together to Save the Barataria and Terrebonne Basins*, Barataria-Terrebonne National Estuary Program, Thibodaux, Louisiana
- Spencer, L (1989) *Winning through Participation*, Kendall/Hunt, Dubuque, Iowa
- Watts, J and Cheramie, K (1995) 'Rallying to save Louisiana wetlands' in Troxel, J (ed) *Government Works: Profiles of People Making a Difference*, Miles Rivers Press, Alexandria, Virginia

INDEX

Page numbers in *italics* refer to figures or boxes; terms in **bold** will be found in the glossary

- aborigines, 127
- abortion, 37
- adaptive development**, 158, 174–80, 215
- Aedes aegypti*
 - biological control outside Vietnam, 194–5
 - biological control in Vietnam, 192–4
 - eradication from Brazil, 186
 - natural predators, 189–90
 - transmission of dengue virus, 185
- Aedes albopictus*, 185
- Africa, 16, 39, 86, 187
- ageing, 54
- ageing populations, 32
- agricultural ecosystems**, 61, 67–9, 217
 - coevolution with social systems, 100–4
 - effects of industrial revolution, 101–4
 - efficiency, 113
 - human inputs, 68–9, 103, 104, 113–14
- Agricultural Revolution**, 27–8, 131, 215
- agriculture
 - Green Revolution, 33–5, 39–40
 - hillside, 116, 142
 - intensification, 28–9, 30, 115–16
 - mechanization, 100–3
 - organic**, 69, 104, 172, 220
 - pollution, 202–3
 - traditional systems, 69, 76, 92–4
- agroforestry**, 94, 215
- Ainu, 128
- albatross, 4
- albedo, 65
- algal blooms, 202
- 'alive' state, 52–4
- alligators, 199
- alternative energy, 178
- Americas *see* Latin America; North America
- Americorps and Delta Service Corps, 212
- anchovy fisheries, 87
- ancient civilizations
 - fall, 145–6, 149–51, 170
 - rise, 147–9
- animals
 - ecosystem consumers, 108–9, 112, *114*, 216
 - religious attitudes to, 127–8, 129
 - social behaviour, 36
 - see also* livestock
- animism**, 126–8, 215
- annual** plants, 78, *138*, 139, 215
- Anopheles*, 189–90
- anti-malarial drugs, 99, 190
- apes, social behaviour, 36
- aquaculture, 196–7
- Aquarium of the Americas, 212
- aquatic ecosystems
 - absorption of wastes, 117
 - pollution, 8, 202
 - see also* Barataria-Terrebonne estuary
- arms race, 18
- Asia
 - agriculture, 117
 - dengue hemorrhagic fever, 186, 187, 194–5
 - food scarcity, 39
 - migrations, 140
 - see also named countries*
- atmospheric pollution, 143–4
- atrazine, 202
- Australian aborigines, 127
- automobile pollution, 143–4
- autonomous** action, 215

230 *Human Ecology*

- Babylon, 145
Bacillus thuringiensis, 187
 bald eagle, 203
 Barataria-Terrebonne estuary, 197–203
 description, 197–9
 environmental problems, 200–3, 211
 history, 199–200
 Barataria-Terrebonne National Estuary
 Program, 203–14
 achievements, 210–11
 action plans, 208–10
 education programs, 212
 facilitation, 213
 goals, 208
 mission statement, 203–4
 openness and inclusion, 213
 plan implementation, 207
 planning workshops, 204–7, 212
 staff, 213
 Barataria-Terrebonne Wildlife Museum, 212
 'benign' nature, 123
 Betamax videorecorders, 18–19
 biogas generators, 7, 12
biological community, 215
 assembly, 47–51, 60, 72–3, 216
 changes, *see* ecological succession
 hierarchical organization, 43–5
biological control
 defined, 189, 215
 of dengue hemorrhagic fever, 189–95
biological production
 human share of, 27, 34, 35
 process, 106, 107, 108, 215
biome, 74, 215
biophilia, 146, 215
 bird territories, 35–6
 birds, 4–5, 155, 203
 birth control, 37, 40
 birth rates
 developing world, 32–3, 40
 industrialized world, 31–2
 Black Plague, 29
 body
 state, 52–4
 temperature regulation, 20
brackish water, 198, 215
 brainstorming, 204
 Brazil
 Aedes aegypti eradication, 186
 human migration, 140
 breast feeding, 37
 brown pelican, 203
 Buddhism, 129
 buffalo, 138–9
 bureaucracies, 46–7
 California, 166–8
 Calvinism, 132
 canals, 200
canopy, 78–80, 82, 215
 capital
 natural, 115, 220
 portable, 142–3
 capitalism, 133, 142–3
 'capricious' nature, 124, 125
 carbon chains, 111–12
 carbon dioxide emissions, 143–4, 178–9
 caring, 175
carrying capacity, 22–4, 35, 215
 Earth, 16, 28, 30
 and Industrial Revolution, 33
 of nations, 38
 case studies, 183
 Barataria-Terrebonne National Estuary
 Program, 196–214
 dengue hemorrhagic fever, 184–96
 things to think about, 214
 cats, 44
 cedars, 72
 CFCs *see* chlorofluorocarbons
 chain of effects, things to think about,
 12–13
 chains of effects, 3–7, 122–3
chaparral, 166, 216
 charcoal production, 89
 children
 contact with nature, 146–7, 165–6
 infanticide, 37
Chilomonas, 190
 Chinese religions, 130
 chlorofluorocarbons (CFCs), 178
 Christianity, 131–2
 cities, 69–70
 community organizations, 167–8, 175–6
 growth, 102, 142, 148–9, 165
 nature within, 145–6, 165–8
 succession, 83
 supply zones, 165
 zone of influence, 148–9, 223
 civilizations, past, *see* ancient civilizations
clear-cutting, 119, 142–3, 216
climax community
 defined, 78, 216
 disturbance, 81–2

- formation, 80–1
- shii* and *kashi* forest, 78–80
- closed-access resources, 144–5, 159, 161–4
- coadaptation**, 61–2, 137, 216
 - people and mosquitoes, 97–9
 - things to think about, 105
- coevolution**, 61–2, 97, 216
 - social system and agricultural ecosystems, 100–4
 - things to think about, 105
- coexistence, 165–8
- Cold War arms race, 18
- colonialism, 46
- common-property resources
 - conflict over, 40
 - overexploitation, 143–5
 - scale of use, 164–5
 - sustainable use, 88–9, 158–64
- community assembly**, 47–51, 60, 72–3, 216
- community education, 177, 179
- community gardening, 177
- community organizations, *see* local community organizations
- community vision, 176–7, 180
- competition, 18–19
 - see also* ecological competition
- complex adaptive systems**, 42, 216
 - things to think about, 59
- complex societies
 - decline of, 149–52
 - growth of, 147–9
- complex system cycles, 56–8, 80–2, 173–4
- conflict, and resources, 40, 69
- Confucianism, 130
- conservation areas, 165–8
- conservation organizations, 155, 179
- consumer** (ecosystem), 108–9, 112, 114, 216
- consumption** (ecosystem), 11, 107–9, 114, 216
- consumption (material), 11, 32, 133, 174–5
- contraception, 37, 40
- controlled burning**, 91–2, 99–100, 216
- cooking fires, 98
- cooking fuel, 5–7
- copepods, 189, 190, *see also* *Mesocyclops aspericornis*
- corporations, 103, 164, 177–8
- counterintuitive** behaviour, 44–5, 92, 216
- cow dung fuel, 6–7
- coypu (nutria), 199, 201, 203
- Creator of the Universe, 126
- crops
 - first developed, 27–8
 - genetic modification, 34
 - Green Revolution, 33–5
 - high-yield varieties, 33, 103, 219
 - monocultures, 101, 102–3, 219
 - polycultures, 21, 69, 101
- cypress trees, 72, 200
- Daisy World, 65–6
- DDT, 99, 187
 - pest resistance, 99, 187
 - pollution, 203
- 'dead' state, 53–4
- death rates, 31
- decomposers**, 67–8, 108, 109, 117, 216
- decomposition**, 67–8, 108, 109, 117, 216
- deer populations, 21, 23, 35
- deforestation
 - cooking fuel collection, 5–7
 - denial, 46
 - and migration, 140–1
 - North Korea, 8
 - soil erosion, 119, 146
- democracy, 175–6
- demons, 129
- dengue hemorrhagic fever
 - biological control outside Vietnam, 194–5
 - biological control in Vietnam, 192–4
 - history, 185–8
 - symptoms, 184
 - transmission, 185
- dengue virus, 184–5, 188
- denial**, 45–6, 216
- desertification**, 85–6, 116, 216
- developing world
 - carbon dioxide emissions, 178–9
 - population growth, 32–3, 40
 - urban growth, 38, 70, 142
- diminishing returns**, 149, 216
- dispersal**, 72–3, 217
- dissolution, 56
- division of labour**, 148, 217
- domestication, 27–8
- 'Dream Time', 127
- drift net**, 3–4, 217
- drug-resistance, 173
- dung, as fuel, 6–7
- Earth
 - carrying capacity, 16, 28, 30

232 *Human Ecology*

- energy input-output, 113
- Earth Summit, 179
- Eastern religions, 128–30
- ecological assessment, 174
- ecological competition**, 49–50, 82–3, 84–5, 217
- ecological niche**, 49–50, 57, 217
- ecological succession**, 77–8, 217
 - as a complex system cycle, 80–2
 - grass ecosystem, 82–3
 - shii* and *kashi* forest, 78–80
 - things to think about, 95
 - urban ecosystem, 83
- ecology**, 1, 217
- economic growth, 175
- economic institutions, 136
- economy of scale**, 102, 217
- ecosystem**, 1, 217
 - community assembly, 47–51, 60, 72–3, 216
 - complex system cycles, 57
 - demands on, 10–11, 115–18
 - design, 62–4
 - emergent properties, 44–5
 - energy flow in, 111–14
 - homeostasis, 63–5, 91
 - natural, 61, 66–7
 - redundancy, 63, 169, 221
 - resilience, 63, 169–74
 - spatial hierarchies, 73, 74
 - state, 55, 64–5, 87–8, 123, 124, 217
 - see also* agricultural ecosystem; urban ecosystem
- ecosystem inputs**, 113–14, 217
 - agricultural ecosystems, 68–9, 103, 104, 113–14
 - and resilience, 169–71
 - urban ecosystems, 113–14
- ecosystem outputs**, 68–70, 217
- ecosystem services**, 2, 114–15, 217
 - intensity of use, 10–11, 115–18
 - ownership/control of, 175
- ecosystem state**, 55, 64–5, 87–8, 217
 - and perceptions of nature, 123, 124
- education, 177, 179
- 'egg-trap effect', 192–3
- emergent properties**, 42, 43–5, 218
- endangered species**, 18, 218
- energy, laws of thermodynamics, 109–10
- energy flow**, 2–3, 106–7, 108–9, 218
 - in ecosystems, 111–14
 - pot of water metaphor, 110–11
- environmental refugees**, 39, 218
- environmental technology
 - dengue biological control, 189–96
 - things to think about, 214
- equilibrium, 56
- estuary**, 196–7, 218
 - see also* Barataria-Terrebonne Estuary
- Europe
 - colonization of America, 132, 138–9
 - population history, 29–30
- eutrophication**, 202
- evolution, 51–2
- exponential population growth**, 14–16, 17, 218
- extended economic zones**, 40, 152, 159
- facilitation processes, 213
- fallow**, 93, 218
- farms
 - size, 102
 - see also* agricultural ecosystems
- feedback, 14, 52
 - in ecological succession, 82–3
 - negative, 14, 20–1, 23, 36–7, 52, 118–19, 220
 - population regulation, 23
 - positive, 14, 17–19, 22–3, 52, 148–9
 - practical significance, 24–5
 - things to think about, 25
- feng-shui, 130
- fertilizers, 34, 39, 103, 104, 116, 171
- fire *see* forest fires
- fisheries succession**, 86–7, 116, 119, 218
- fishing
 - destruction of marine mammals, 3–5, 155
 - mercury contamination, 8
 - supply and demand control, 119
 - sustainable, 119, 160–1
 - see also* overfishing
- flood control, 171–2
- floods, 8–9
- food chain**, 8, 218, *see also* food web
- food chain efficiency**, 112, 218
- food scarcity, 39
- food security, 173
- food web**, 60, 218
 - energy flow in, 109–11, 112–13
 - energy losses, 108, 110–11, 112
 - island community, 49–50
 - see also* food chain
- forest fires
 - controlled burning, 91–2, 99–100

- ecological functions, 90
- protection, 45, 90–2, 172
- in swidden agriculture, 93
- forest management
 - sustainable, 88–90, 142, 161–4
 - unsustainable, 115–16, 119, 142–3
- forests
 - ecological functions, 9
 - ecological succession, 78–80
 - migration to, 140–1
 - see also* deforestation; forest fires; forest management
- fossil fuel energy, 68–9, 114, 170–1
- 'fragile' nature, 123–4
- Francis of Assisi, Saint, 131
- free market economy, 142–3
- fur trade, 199

- Gaia hypothesis, 64–5
- gardening, community, 177
- genetic mutations, 51–2, 99
- genetically modified crops, 34
- girls, infanticide, 37
- global economy, 164
- global warming, 143–4, 173, 178–9, 188
- goats, 85, 146
- God, 130–1
- government
 - control of resources, 165
 - lobbying of, 167–8
 - role in adaptive development, 179
- grasslands
 - ecological competition, 84–5
 - ecological succession, 82–3
 - overgrazing, 84–6, 116, 117
 - see also* Great Plains
- Great Plains, 138–9
- Green Revolution, 33–5, 39–40
- greenhouse gas emissions, 143–4, 178–9
- guardian spirits, 128
- Gulf of Mexico, 197, 198–9

- habitat**, 101, 218
- heat loss, ecosystems, 108, 110–11, 112
- heating fuels, 170–1
- herbicides, 202
- hierarchical organization**, 43–5, 73, 74, 218
- high-yield varieties** (crops), 33, 103, 219
- hillside agriculture, 116, 142
- Hinduism, 128–9
- homeostasis**, 219

- ecosystem, 63–5, 91
- organism, 20
- Homo erectus*, 26
- Homo habilis*, 26
- Homo sapiens*, 26
- housing
 - heating fuel, 170–1
 - and malaria, 98–9
 - public (example of counterintuitive behaviour, 44–5
- human cultures
 - evolution, 52
 - and perceptions of nature, 121–2
- human ecology**
 - defined, 1–5, 219
 - things to think about, 12–13
- human migrations, 39, 137–41
- human population, 16
 - history, 26–35
 - and quality of life, 38–40
 - social regulation, 35–8
 - things to think about, 41
- human social systems *see* social systems
- human values, 174–5
- human–ecosystem interactions *see*
 - sustainable human–ecosystem interaction; unsustainable human–ecosystem interaction
- human-induced succession**, 76, 83–4, 116–17, 219
 - aquatic ecosystems, 117
 - desertification, 85–6, 116, 216
 - fisheries, 86–7, 116, 119
 - forest fire protection, 90–2
 - forest management, 88–90, 119
 - okay/not okay principle, 87–8
 - pasture degradation, 84–5, 116, 117
 - traditional agriculture, 92–4
- hunger, 39
- hunter-gatherers, 27, 122, 127–8
- hydroelectric dams, 171

- India
 - deforestation and cooking fuel, 5–7
 - Hindu religion, 128–9
- Indonesia, 94, 140
- Industrial Revolution**, 30–3, 219
 - effects on agriculture, 30–1, 100–2
 - human population, 31–3
 - and use of ecosystem services, 117
- industrialized nations
 - carbon dioxide emissions, 178–9

234 *Human Ecology*

- food supply, 173
- material consumption, 11, 32, 133, 174–5
- population growth, 31–2
- infanticide, 37
- information
 - distortion, 45
 - transfer, 2–3
- irrigation agriculture, 33
 - and deforestation, 6
 - Middle East failure, 145–6
 - salinization, 116–17
- Islam, 132–3
- islands, community assembly, 48–51, 57
- Japan
 - Minamata disease, 8
 - natural forest ecosystems, 71–2, 78–80
 - population ageing, 32
 - religion, 128
 - traditional forest management, 88–90, 161–4
- Japanese Zen proverb, 154
- Javanese polyculture, 94
- Judaism, 130–1
- Kansai region, Japan, 71
- karma, 129
- kebun-talun* cropping, 94
- kelp forests, 5
- killer whales, 5
- Koran*, 132
- Korea, North, 8, 39
- Krishna, 129
- kumi*, 162
- Kyoto Protocol, 178–9
- lagoons, pollution, 84
- Lake Victoria, 16
- land degradation, 39, 84–6, 116–17, 139
- land subsidence**, 200–1, 219
- landscape mosaics**, 60–1, 71–4, 166
- larvicides**, 186, 187–8
- Latin America
 - dengue fever, 187
 - fisheries, 87
- leaf litter**, 90–2, 119, 219
- legume**, 62, 219
- levees, 171, 200
- lightning, 82
- livestock
 - domestication, 27–8
 - genetic modification, 34
 - overexploitation, 138–9
 - overgrazing of pastures, 84–6, 116, 117
- lobbying, 167–8
- local communities
 - adaptive development, 175–7
 - ecosystem protection, 167–8
 - role in biological control programs, 193, 195, 196
- local environmental management
 - Barataria-Terrebonne National Estuary Program, 203–14
 - things to think about, 214
- logging
 - excessive, 115–16, 119
 - selective**, 92, 142, 221
- long line fishing, 4–5, 155
- Los Angeles, 167–8
- malaria, 97–9, 187, 189–90
- malathion, 188
- mammals
 - extinctions in America, 137
 - losses of marine, 3–5, 155
 - social behaviour, 36
 - see also* livestock
- mangrove ecosystems, 196–7
- marine mammals, 3–5, 155
- market forces, 141–2
- marriage, 37
- marsh**, 198, 201, 202, 219
- material consumption, 11, 32, 133, 174–5
- material cycling**, 2–3, 106–11, 219
- Mead, Margaret, 181
- medicine, 173
- Meiji Restoration, 164
- mercury pollution, 8
- Mesocyclops aspericornis*, 189
 - biological control of dengue, 192–6
 - distribution, 194
 - mass production, 190–1
 - survival in water containers, 191
- Mesopotamia, 145–6
- methane generation, 7, 12
- metropolitan region**, 73, 74, 219
- mice, 44
- microorganisms, 108–9, 112
- Middle East
 - agricultural revolution, 28
 - collapse of civilizations, 145–6, 170
 - religion, 130–1
- migrations, 39, 137–41
- military, 150–1

- Minamata disease**, 8, 219
 mineral cycling *see* material cycling
mineral nutrients, 62, 90, 93, 108, 109, 112, 219
 Mississippi River, 197, 200, 202–3, *see also* Barataria-Terrebonne Estuary
 Mohammed, 132
 molecules, 44
 monastic traditions, 131
 monkeys, social behaviour, 36
monoculture, 101, 102–3, 139, 219
 moral codes, 125–6
 mosquitoes
 biological control, 192–5
 dengue transmission, 185
 malaria transmission, 97–9
 natural predators, 189–90
 pesticide control, 187–8
 pesticide resistance, 99, 187
 Mother Earth, 126
 mountain regions
 agriculture, 142
 deforestation, 146
 expansion of cities, 166–8
 malaria, 97–9
 migrations, 140
 multinational corporations, 164
 mutations, genetic, 51–2, 99
mycorrhizae, 62, 219
- nail puzzle (stability domains), 54
 Native Americans
 controlled forest fires, 99–100
 European colonization of North America, 132, 138–9
 occupation of Barataria-Terrebonne estuary, 199
 occupation of Great Plains, 138
 religion, 127–8, 133
natural capital, 115, 220
 natural disasters, 176
natural ecosystem, 217
 effects of urban growth, 165
 organization, 61, 66–7
 nature
 childhood contact with, 146–7, 165–6
 common perceptions, 121, 122–5
 dangers of romanticizing, 133–4
 religious attitudes, 128–33
 nature conservation organizations, 179
negative feedback, 14, 20–1, 52, 220
 in ecological succession, 82–3
 human population control, 36–7
 population regulation, 23
 practical significance, 24–5
 in supply and demand, 118–19
 things to think about, 25
 neighbourhoods, 73, 74, 83, 151
 net primary production *see* primary production, net
 NGOs *see* non-governmental organizations
 niche *see* ecological niche
 nitrogen pollution, 202
nitrogen-fixing bacteria, 62, 220
 non-governmental organizations, 179
non-renewable natural resources, 10, 220
- North America
 Aedes aegypti distribution, 185–6
 European colonization, 132, 138–9
 first migrations, 137
 see also Barataria-Terrebonne Estuary; Santa Monica mountains; United States
 North Atlantic fisheries, 153
 nutria (coypu), 199, 201, 203
 nutrient cycling *see* material cycling
nutrient pump, 93, 220
- oak forests
 natural ecology, 71–2, 78–80
 traditional Japanese management, 89–90
 okay/not okay principle, 87–8, 116, 123
 open-access resources, 143–4
organic farming, 69, 104, 172, 220
 organisms, 43
overexploitation, 115–18, 220
 role of wishful thinking, 152–4
 tragedy of the commons, 143–5
overfishing, 220
 fish species changes, 86–7, 116, 119, 218
 new technologies, 141
 prevention, 119, 160–1
 role of wishful thinking, 152–4
 tragedy of the commons, 144
 overgrazing, 84–6, 116, 117
overshoot, 23, 86, 220
 ozone layer depletion, 178
- Paramecium caudatum*, 190
parasite, 61, 220
 paris green, 186
 pastures
 agricultural development, 138–9
 inputs, 68

236 *Human Ecology*

- overgrazing, 84–6, 116, 117
- pathogen**, 61, 220
- perception**, 220
- perceptions of nature, 121–2
 - benign and perverse, 123, 124
 - capricious, 124, 125
 - durable, 124–5
 - and ecosystem management, 122
 - everything is connected, 122–3
 - fragile, 123, 124
 - things to think about, 134–5
 - see also religions
- perennial** plants, 89, 138, 220
- 'perverse' nature, 123
- pest control, biological, 101, 172, 189–96
- pesticide trap, 172–3
- pesticides
 - pest resistance, 99, 172–3, 187
 - pollution, 103, 202–3
 - use, 34, 39, 103, 104, 116, 172–3
- petroleum energy, 68–9, 114
- Phanboi, dengue hemorrhagic fever
 - control, 192–4
- photosynthesis, 106–8, 111–12
- phytoplankton**, 8, 221
- pine forests, 71–2, 80
- plants
 - coadaptation, 61–2
- primary production**, 106–8, 111–12, 221
 - see also crops
- policies (social), 58
- pollution
 - aquatic ecosystems, 8, 84, 202–3
 - atmosphere, 143–4
- polyculture**, 21, 69, 101, 139
- polygamy, 37
- population**, 221
 - complex system cycles, 56–7
 - emergent properties, 43–4
- population growth
 - exponential, 14–16, 17, 218
 - overshoot**, 23, 86
 - see also human population
- population pressure**, 28, 221
- population regulation**, 21–4, 221
 - human social mechanisms, 35–8
 - predators, 35
- portable capital, 142–3
- positive feedback**, 14, 17–19, 52, 221
 - in ecological succession, 82–3
 - population regulation, 22–3
 - practical significance, 24–5
- social complexity and urban growth, 148–9
 - things to think about, 25
- pot of water metaphor, 110–11
- precautionary principle**, 118, 153–5, 157, 221
- predator**, 221
 - of insect pests, 101, 172, 189
 - population regulation, 35, 49–50
- prey, 44, 61
- primary production**, net
 - human control of, 27, 34, 35
 - process, 106–8, 111–12, 221
- private sector, 103, 164, 177–8
- profit motive, 143
- Protestant Reformation, 132
- protozoa, 190
- quality of life, 38–40
- rainforests, 140–1
- redundancy**, 63, 169, 221
- Reed, Walter, 186
- refugees, environmental, 39
- reincarnation, 129
- religions, 122, 125–6
 - Eastern, 128–30
 - modern 'economic', 133
 - moral codes, 125–6
 - spirit, 126–8
 - Western, 130–3
- renewable natural resources**, 106, 221
 - conflict over, 40
 - intensity of use, 10–11
 - ownership/control, 132, 143–5, 159, 160, 165
 - scale of use, 164–5
 - supply and demand, 118–19
 - sustainable use, 88–9, 158–64
- reorganization, 56
- resilience**, 158, 221
 - complex societies, 150
 - ecosystem, 63, 169–74
 - trade off with stability, 170–1
- resources see common property resources; non-renewable resources; renewable natural resources
- respiration**, 108, 111, 112, 221
- rivers, 117, 197, 200, 202–3
- Rockefeller Foundation, 186
- romanticizing, 133–4
- rubber band analogy, 169

- S-shaped (sigmoid) curve, 23–4, 43–4
 sacred groves, 127
 Sahel region, 86
salinization, 116–17, 221
salt-water intrusion, 200, 221
 Santa Monica mountains, 167–8
 sardine fisheries, 86–7
satoyama system, 89–90, 221
 science, 126, 132
 sea otters, 5
 sea urchins, 5
 seabirds, 4–5, 155
 seals, 5
selective logging, 92, 142, 221
 self-organization, 47–52
 biological communities, 47–51
 human social systems, 51–2
 landscape mosaics, 73–4
 sewage pollution, 202
 sewage treatment, 211
 shifting cultivation *see* swidden agriculture
shii and *kashi* forest
 natural succession, 78–80
 traditional management, 88–90
 Sierra Club, 167–8
 slash-and-burn agriculture *see* swidden agriculture
 social complexity, 147–52
 costs of, 149–52
 growth of, 147–9
social institutions, 137, 145, 221
 sustainable resource use, 158–60
 village forest groups, 161–4
social organization, 51–2, 221
 see also social complexity
social systems, 1–2, 222
 coevolution and coadaptation, 97
 complex system cycles, 57–8
 counterintuitive behaviour, 44–5
 emergent properties, 45–7
 feedback, 18, 20–1, 24–5
 self-organization, 51–2, 221
 states, 55
 traditional, 97, 133–4
 see also social complexity
soil erosion, 222
 agriculture, 85–6, 102, 116, 139, 142
 deforestation, 119, 146
 soil fertility, 67–8, 171
 soil salinization, 116–17
 South Pacific islands, lagoon pollution, 84
 Soviet Union, former, 18
 Spanish moss, 199
Spartina grass, 203
 species, endangered, 18
 spirit religions, 122, 126–8
 Spirit Who Controls the Land, 126
stability, 20, 170–1, 222
stability domains
 defined, 52–5, 222
 fisheries, 153
 human-induced succession, 87–8
 and perceptions of nature, 123, 124
 resilience, 169
 states, 52–5
 ecosystem, 55, 64–5, 87–8, 123, 124, 217
 human body, 52–4
 social systems, 55
subsistence farming, 100–1, 222
 subsistence living, 147–8
 succession, *see* ecological succession;
 human-induced succession
 sunlight, 27, 65–6, 111, 113
 supermarkets, 103
 supply and demand, 118–19
 sustainable development, 7–9
sustainable human–ecosystem
 interaction, 157–8, 222
 adaptive development, 174–80
 common property resource use, 158–65
 complex system cycles in, 173–4
 nature in cities, 165–8
 resilience, 169–74
 things to think about, 13, 181–2
swamp, 198, 201, 222
swidden agriculture, 92–4, 222
 ‘switching’, 52–4
symbiosis, 61–2, 222
 Taiwan, dengue control, 187
 Taoism, 130
 technology
 agricultural, 28–9, 30
 resource overexploitation, 141–2
 resource use, 10–11
 television set, 62–4
 temperature regulation
 Earth surface, 65–6
 human body, 20
 territoriality, 36–7, 38, 137, 159
 Thailand, 186
 thermodynamics, laws of, 109–10
 things to think about
 chain of effects, 12–13

238 *Human Ecology*

- coadaptation and coevolution, 105
- complex adaptive systems, 59
- ecological succession, 95
- ecosystem organization, 75
- ecosystem services, 119–20
- environmental technology, 214
- feedback systems, 25
- human population explosion, 41
- local environmental management, 214
- perceptions of nature, 134–5
- sustainable human–ecosystem
 - interaction, 13, 181–2
- unsustainable human–ecosystem
 - interaction, 155–6
- Tokugawa period, 163–4
- Topanga State Park, 168
- traditional agriculture, 69, 76, 92–4
- traditional forest management, 88–90, 161–4
- traditional social systems, 97, 133–4
- tragedy of the commons**
 - prevention, 158–65
 - why it occurs, 143–5, 222
- trash fish**, 86–7, 119, 222
- tropical forests, 140–1
- Turkey, coastal fisheries, 160–1
- turtles, 3, 155
- United Nations Conference on Environment and Development, 179
- United States
 - Aedes aegypti* distribution, 185–6
 - arms race, 18
 - colonization, 132, 137–9
 - forest fire protection, 90–2
 - public housing, 44–5
 - see also Barataria-Terrebonne Estuary; Santa Monica mountains
- United States Forest Service, 90–2
- unsustainable** human–ecosystem
 - interaction, 136–7, 222
 - migrations, 137–41
 - new technologies, 141–2
 - things to think about, 155–6
- urban ecosystems**, 217
 - Aedes aegypti* survival, 193–4
 - alienation from nature, 146–7
 - coexistence with nature, 165–8
 - collapse of, 149–52
 - growth, 147–9
 - human energy inputs, 113–14
 - organization, 61, 68, 69–70
 - spatial hierarchies, 73, 74
 - succession, 83
- urbanization, 38, 70, 136, 147–9, 164
- values**, 174–5, 222
- VHS videorecorders, 18–19
- videorecorders, 18–19
- Vietnam
 - dengue hemorrhagic fever, 188, 194
 - dengue hemorrhagic fever control
 - program, 192–4
 - malaria, 97–9
- village forests, Japan, 161–4
- Vishnu, 129
- vision, 176, 204–5, 212
- wars, 40
- wastes, 106, 115, 117
- water, conflict over, 69
- water containers
 - dengue fever control, 189–96
 - dengue fever transmission, 185
- water hyacinth, 14–16, 202
- watershed**, 9, 222
- wealth, 7, 11, 132
- Western religions, 130–3
- wetlands**, 198, 200–1, 223
- whales, killer, 5
- wishful thinking, 152–4
- World Council of Churches, 132
- World Health Organization, 99, 187
- World War II, 186
- worldview**, 121, 133, 223
- yellow fever, 184, 185–6
- yin-yang, 130
- Zen proverb, 154
- zone of influence**, 148–9, 223
- zooplankton**, 8, 223